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PALAEOMAGNETISM AND K–Ar AGES OF TRIASSIC IGNEOUS ROCKS FROM THE ISCHIGUALASTO-ISCHICHUCA BASIN AND PUESTO VIEJO FORMATION, ARGENTINA

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The palaeomagnetism of Middle Triassic (224 ± 5 m.y.) igneous rocks from the Ischigualasto-Ischichuca Basin (67°40'W, 30°20'S) was investigated through 86 oriented hand samples from 11 sites. At least one reversal of the geomagnetic field has been found in these rocks. Nine sites yield a palaeomagnetic pole at 239°E, 79°S (α_{95} = 15°, k = 13).

The K–Ar age determinations of five igneous units of the Puesto Viejo Formation give a mean age of 232 ± 4 m.y. (Early Triassic). The palaeomagnetism of six igneous units of the Puesto Viejo Formation (68°W, 35°S) was investigated through 60 oriented samples. These units, two reversed relative to the present magnetic field of the Earth and four normal, yield a pole at 236°E, 76°S (α_{95} = 18°, k = 14).

Data from the Puesto Viejo Formation indicate, for the first time on the basis of palaeomagnetic and radiometric data, that the Illawarra Zone, which defines the end of the Kiaman Magnetic Interval, extends at least down to 232 ± 4 m.y. within the Early Triassic. The palaeomagnetic poles for the igneous rocks of the Ischigualasto-Ischichuca Basin and Puesto Viejo Formation form an "age group" with the South American Triassic palaeomagnetic poles (mean pole position: 239°E, 77°S; α_{95} = 6.6°, k = 190). The Middle and Upper Permian, Triassic and Middle Jurassic palaeomagnetic poles for South America would define a "time group" reflecting a quasi-static interval (mean pole position: 232°E, 81°S; α_{95} = 4°, k = 131).

1. Introduction

Valencio and Mitchell [1] have recently defined the Late Palaeozoic Interval of reversed geomagnetic polarity (Kiaman Magnetic Interval [2]) as that in which the geomagnetic field had predominant reversed polarity from the Moscovian-Bashkirian (Debal'tseva Zone) at the base to either the Upper Permian or the Lower Triassic at the top (Illawarra Zone). The evidence is not conclusive about the age of the period of intermittent reversals of the geomagnetic field (Illawarra Zone) which define the end of the Kiaman Magnetic Interval. The only palaeomagnetic-radiometric data available within or close to the Illawarra Zone are those from the Manicovagan dacite (normal remanence, 225 ± 3 m.y. [3]), of Middle Triassic age according to the time scale of Lambert [4]. In the western and northwestern part of Argentina are exposed igneous rocks (Ischigualasto-Ischichuca Basin and Puesto Viejo Formation) which has been assigned to the Early Triassic. This report presents the data obtained from palaeomagnetic and radiometric studies of some of these igneous rocks and an interpretation of them in terms of the Illawarra Zone.

It has been indicated that more South American palaeomagnetic poles are necessary to define the mean Triassic polar position precisely [5]. The palaeomagnetic poles computed for the igneous rocks from the Ischigualasto-Ischichuca Basin and Puesto Biejo Formation presented here allow us to define the South American mean Triassic polar position more precisely.

2. Geological setting and sampling

2.1. Ischigualasto-Ischichuca Basin

In tne central east part of the Province of San Juan, close to the border with the Province of La Rioja, Ar-

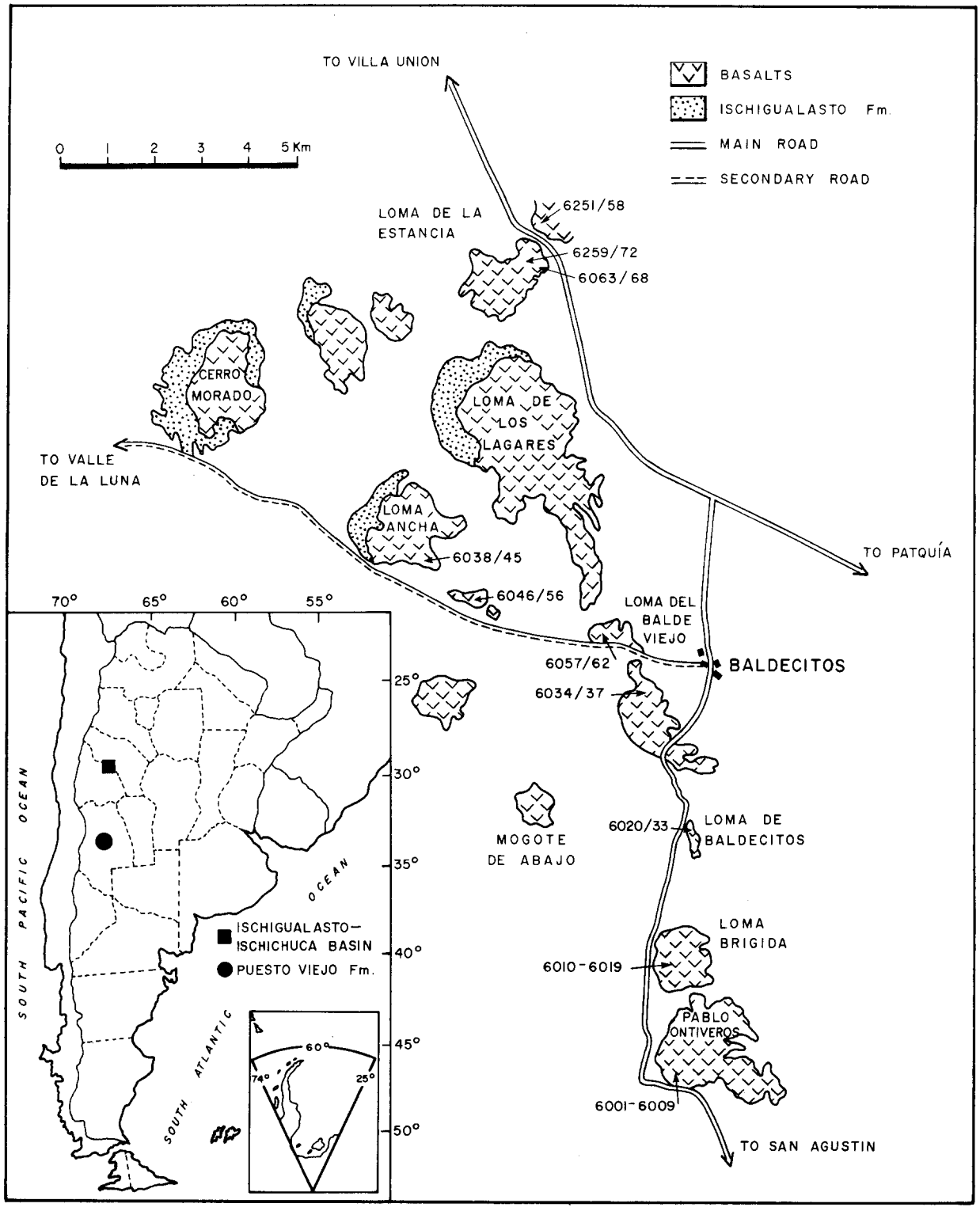


Fig. 1. Ischigualasto-Ischichuca Basin. Geologic sketch map of the area close to Baldecitos showing location of palaeomagnetic sample sites (according to the photogeological chart of Sierras Pampeanas). The positions of the Ischigualasto-Ischichuca Basin and exposures of Puesto Viejo Formation are shown in the reference map.

gentina, are exposed continental deposits and igneous rocks included in Los Rastros and Ischigualasto Formations [6], of the Ischigualasto-Ischichuca Basin (Fig. 1). These continental deposits are characterized by the presence of excellent reptile fossil assemblages. On the basis of these fossils, Bonaparte [7] has suggested a Ladinian age (Middle Triassic) for Los Rastros Formation and a Carnian age (Upper Triassic) for the Ischi-

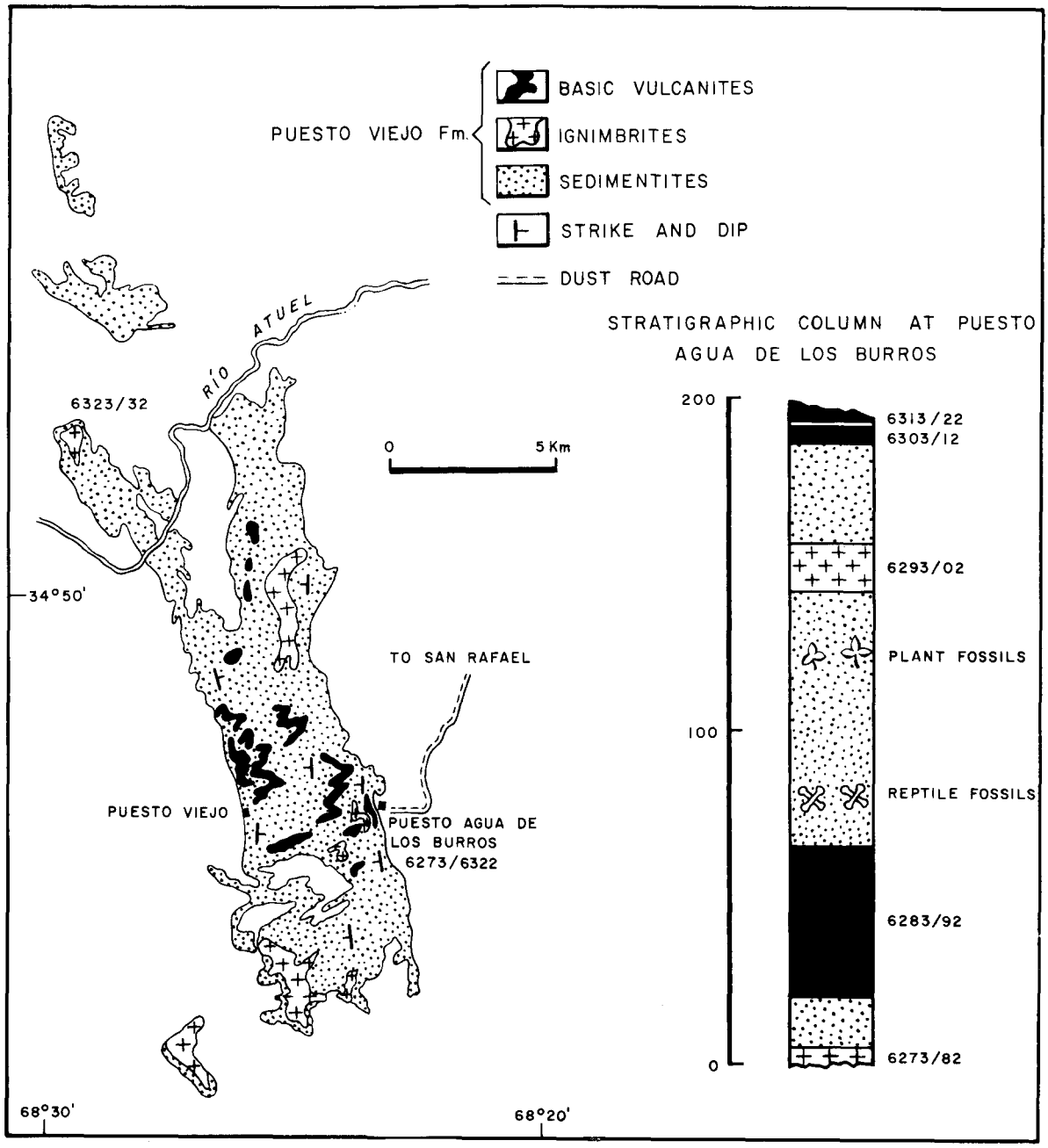


Fig. 2. Puesto Viejo Formation. Schematic geologic map and simplified geological section at Puesto Agua de los Burros according to González Díaz [11]. Sites where the igneous rocks of the Puesto Viejo Formation were collected are also shown. The numbers refer to the samples collected at each site.

gualasto Formation. It has been suggested that the basic igneous rocks of the Ischigualasto-Ischichuca Basin are of intrusive origin and related with the Early Cretaceous magmatic events of the Province of Cordoba (Argentina) and Serra Geral (Brazil) [8,9]. Andreis [10] carried out a detailed study of the basic igneous exposures of the Cerro Guandacol of that basin, which had been previously considered as intrusives, showing their volcanic origin. On the basis of stratigraphic evidence that author suggested a Lower Triassic age for the igneous exposures of the Cerro Guandacol.

The igneous rocks of the Ischigualasto-Ischichuca Basin were sampled at eight different sites (Fig. 1). Eighty-six oriented hand samples were collected close to the locality of Baldecitos, at about 67°40′ longitude west and 30°20′ latitude south. In this area the basaltic rocks rest conformably on the continental sediments. The outcrops of the basaltic rocks are generally small and cover the top of scattered low hills (Lomas).

At Loma Ancha and Loma del Balde Viejo the igneous rocks rest on nearly horizontal poorly indurated yellowish sandstones. At Loma de la Estancia the basaltic rocks rest on well stratified light-gray sandstones which dip gently (less than 10°) to the east. At Loma de la Estancia are exposed several igneous units at different stratigraphic levels. Twenty-seven hand samples were collected from the lower, middle and higher levels at this site.

2.2. Puesto Viejo Formation

In the central part of the Province of Mendoza, Argentina, close to San Rafael city, at about 68° longitude west and 35° latitude south, are exposed continental deposits included in the Puesto Viejo Formation (Fig. 2) [11]. These deposits, about 300 m thick, consist of conglomerates, sandstones, argilites and limestones intercalated with vulcanites of basic (basalts and andesites) and acid (ignimbrites) composition. The deposits of the Puesto Viejo Formation rest with erosional unconformity on the eruptive rocks of the Cerro Carrizalito Group (Choiyoilitense, Middle Permian, 263 ± 5 m.y. [1]). The Cerro Carrizalito Group is underlain by the Cochicó Formation (Lower Permian, 276 ± 14 m.y. [12]). Palaeomagnetic studies of the lava flows of the Cerro Carrizalito Group and its equivalent Cerro Bola and Cerro Colorado Formations were carried out by Valencio [13] and Vilas [14], respectively.

The sandstones of the Puesto Viejo Formation are characterized by the presence of reptile fossil assemblages (therapsids of the Dicynodontia and Cynodontia groups). On the basis of these fossils González Díaz [11] has suggested an age not younger than Middle Triassic for the Puesto Viejo Formation. Bonaparte [7] has suggested an Anisian age for the Puesto Viejo Formation based on the "strong affinities" of the therapsids occurring at the sandstones of this formation with African forms of Scythian and Anisian age. Recently this author (personal communication) has suggested a Lower Triassic age for the upper section of the Puesto Viejo Formation.

The volcanic units of the basal section of the Puesto Viejo Formation were sampled at two different localities, Puesto Agua de los Burros where the strata dip gently (10° to 15°) to the west (Fig. 2) and Rio Atuel Canyon where the strata are horizontal. At Puesto Agua de los Burros (Quebrada de los Fosiles) five volcanic units are exposed, the lowest of which is an ignimbrite 3 m thick overlain by 20 m of conglomerates and sandstones (Fig. 2). In the Rio Atuel Canyon is exposed only one volcanic unit (ignimbrite) 3 m thick equivalent to the lowest volcanic unit of Puesto Agua de los Burros. Ten oriented hand samples were collected from each of the six volcanic units.

3. K–Ar ages

3.1. Ischigualasto-Ischichuca Basin

The radiometric age of the igneous rocks from the Ischigualasto-Ischichuca Basin outcropping near Baldecitos village is known from the studies of González and Toselli [15]. These authors reported ages near 224 m.y. (mean 224 ± 5 m.y.) for a basalt and a diabase of this zone. This Middle Triassic age, according to the time scale of Lambert [4], is thought to be representative of the igneous rocks of the Ischigualasto-Ischichuca Basin outcropping close to Baldecitos.

3.2. Puesto Viejo Formation

Samples from the six igneous units of the Puesto Viejo Formation collected at Puesto Agua de los Burros and Rio Atuel Canyon were dated by the K–Ar method at the Instituto de Geocronología y Geología

TABLE 1

Analytical data of the radiometric age determinations for the igneous rocks from the Puesto Viejo Formation, Province of Mendoza, Argentina

Unit	Sample No.	K (%)	^{40}Ar rad (10^{-10} mol/g)	^{40}Ar atm (%)	Age (m.y.)
2 A. Burros, basalt	6305/6	3.54	16.4	18.2	232 ± 10
3 A. Burros, ignimbrite	6302	2.49	11.18	8.8	236 ± 10
4 A. Burros, basalt	6283/92	1.72	7.54	32.5	232 ± 10
5 A. Burros, ignimbrite	6275	5.07	22.15	4.1	230 ± 10
6 R. Atuel, ignimbrite	6330	3.56	15.44	20.6	228 ± 10

Isotópica, Argentina. The argon extraction was accomplished using the isotope dilution method. The potassium analysis was performed in duplicate for each sample. Linares et al. [16] have described the equipment and techniques used in both determinations.

The K–Ar ages and the analytical errors were calculated according to Cox and Dalrymple [17]. We were not able to obtain reliable results of the andesite from Agua de los Burros. The analytical data of the K–Ar determinations for the other five units from the Puesto Viejo Formation are given in Table 1; the ages obtained for these igneous units are concordant. It seems improbable that the processes of weathering and alteration operating on the samples collected from different units would lead to argon loss of precisely the same amount and, accordingly, the mean of the ages is believed to provide a valid estimate of the age of the Puesto Viejo Formation. Therefore, the radiometric study suggests a mean age of about 232 ± 4 m.y. (p = 0.05) for the igneous rocks from the Puesto Viejo Formation. This Early Triassic age according to the time scale of Lambert [4] is thought to be representative of the Puesto Viejo Formation.

4. Palaeomagnetic study

All hand samples were oriented by magnetic and sun compass. In most cases magnetic azimuths of the samples agree with sun-compass azimuths within ±3°. Where the disagreement was more than 3° the sun-compass azimuths were used.

TABLE 2

Summary of the palaeomagnetic data for the lava flows from the Ischigualasto-Ischichuca Basin, Province of San Juan, Argentina

Site	Sample No.	Direction of natural remanent magnetization					Direction of cleaned remanent magnetization					Virtual geomagnetic pole				
		N	D (°)	I (°)	k	α_{95} (°)	N	D (°)	I (°)	k	α_{95} (°)	lat. S (°)	long. E (°)	$d\psi$ (°)	$d\chi$ (°)	sym-bol
1 Loma Brigida	6010/19	10	203	54	12	14	10	195	59	10	16	74	245	18	24	●
2 Loma Baldecites	6020/33	14	325	−10	2	43	13	198	42	7	18	73	187	14	22	◆
3 Dust road (Baldecitos-Valle de la Luna, km 2)	6034/37	3	209	72	57	17	3	223	73	29	23	49	260	38	42	▼
4 Loma Ancha	6038/45	7	32	63	2	65	5	183	64	89	8	74	283	9	12	▲
5 Dust road (km 6)	6046/56	11	321	15	1	89	10	198	58	8	18	73	237	19	26	■
6 Loma Balde Viejo	6057/62	6	128	14	1	n.c.	5	207	28	14	21	61	178	13	23	◆
7 Loma de la Estancia																
7a Lower section	6063/65; 6251/54	7	348	−39	46	9	4	342	−31	61	12	69	57	7	13	▽
7b Lower section	6066/68; 6255/58	7	348	26	2	80	5	190	73	21	17	61	281	26	30	▼
7c Middle section	6259/66	8	358	−37	15	15	8	352	−30	14	15	75	87	9	17	△
7d Higher section	6267/71	5	188	−20	1	n.c.	–	–	–	–	–	–	–	–	–	–
8 P. Ontiveros	6001/08	8	344	−24	1	68	–	–	–	–	–	–	–	–	–	–

After being returned from the field, the samples were cut into cylinders 1 cm in length and 2.5 cm in diameter; four cylinders were cut from each sample. A spinner magnetometer was used to measure the natural remanent magnetization of these cylinders. One cylinder from each sample was subjected to a.c. demagnetizing fields increasing in steps of 50 Oe up to or above 300 Oe peak field. In this way the best a.c. demagnetizing field for isolating the stable remanent magnetization of each sample was determined. The criterion used for the choice of the best demagnetizing field strength was that the direction of the remanence of the pilot cylinder exhibited no further change in response to higher demagnetizing field strengths. The other three cylinders of the same hand sample were submitted to this best demagnetizing field for a better definition of the stable remanent magnetization of the sample.

Fisher's statistics [18] was used to compute the mean direction of the stable remanence of each sample on the basis of the direction of the cleaned remanent magnetization of its cylinders and of each palaeomagnetic site or geological unit on the basis of the stable remanence of its samples.

4.1. Ischigualasto-Ischichuca Basin

The mean directions of NRM of each palaeomagnetic site in Ischigualasto-Ischichuca Basin are given in Table 2. It can be seen that the within-site precision is good for the sites 1, 3, 7a and 7c, with k values as high as 57 for site 3. On the other hand the within-site precision is bad for the other sites with k values as low as 1 for sites 5, 6, 7d and 8.

Alternating-field partial demagnetization curves of representative samples from each site are shown in Fig. 3. It can be seen from the shapes of the demagnetization curves that a wide spectrum of remanent coercivities is represented. The best demagnetizing field strength for the samples from sites 2, 4, 5, 6 and 7 is about 150 Oe. The best demagnetizing field strengths for sites 3 and 1 are 50 and 200 Oe respectively. We obtained reliable results in samples from only nine of the sites (Table 2). It was not possible to isolate stable remanence in samples from site 7d. We isolated stable remanence in samples from site 8, but we doubt whether these samples were "in situ". As the directions of the stable remanence of these samples are badly scattered, we have not included them in

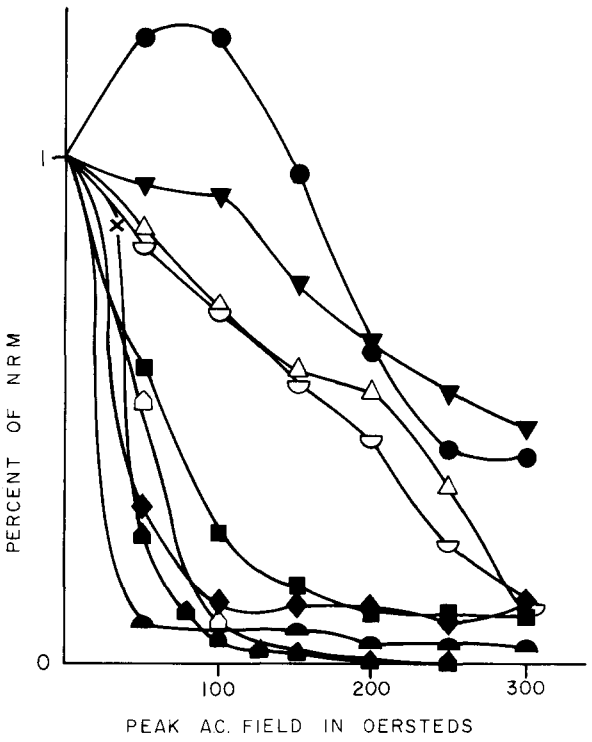


Fig. 3. Alternating-field partial demagnetization curves of representative samples from each site in Ischigualasto-Ischichuca Basin. Symbols as in Table 2.

Table 2. Directions of the stable remanence of samples from each of the nine sites are well grouped. Sites 2, 4, 5, 6 and 7b are substantially better grouped after a.c. demagnetization than before (Table 2). Samples from these sites have reversed stable magnetization; therefore the high scattering of the directions of natural remanent magnetization of samples from these sites could be the result of secondary components directed in the direction of a normal polarity field. These secondary components may be due to viscous remanent magnetization. The shapes of the demagnetization curves of these sites indicate that the most soft components of the remanence have been destroyed during the a.c. demagnetization.

The mean directions of magnetization for each site after a.c. demagnetization are given in Table 2 and plotted in Fig. 4. Seven of the sites yield mean directions toward the south with positive inclinations. These sites are reversed with respect to the present geomagnetic field direction. The occurrence of well-grouped directions, divergent from the present field, and the

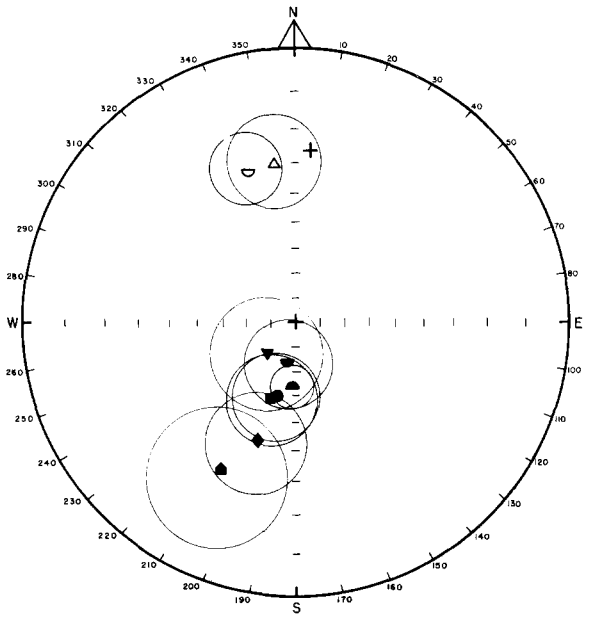


Fig. 4. Site-mean directions of magnetization for the igneous rocks of the Ischigualasto-Ischichuca Basin after a.c. demagnetization (Table 2). Closed symbols are lower hemisphere projections. Circles of 95% confidence are drawn around each site-mean direction. + = geomagnetic field.

presence of reversed magnetization indicate that substantial secondary remanences are absent. The inexact opposition of the normal and reversed site directions could be due to a small secondary component, in which case we could deduce the direction of the larger primary one by forming a population comprising the reversed directions as they stand together with the normal directions turned through 180° (see entry 2 in Table 4). However, the normal magnetized samples come from Loma de la Estancia while most of the reversely magnetized samples were collected at other different sites. Thus, they may have different ages. This interpretation is preferred to the one given above taking into account that all the samples were submitted to a detailed a.c. cleaning.

4.2. Puesto Viejo Formation

The mean directions of NRM, referred to the palaeohorizontal, of each unit in Puesto Viejo Formation are given in Table 3. The within-site precision is quite good for the units 1, 3 and 6, with k values as high 61 for unit 3. On the other hand, it can be seen that the within-site precision for the units 2, 4 and 5 are low.

The a.c. demagnetization curves of representative

TABLE 3

Summary of the palaeomagnetic data and K–Ar ages for the lava flows from the Puesto Viejo Formation, Province of Mendoza, Argentina

Unit	Sample No.	Direction of natural remanent magnetization					Direction of cleaned remanent magnetization					Virtual geomagnetic pole					K–Ar age (m.y.)
		N	D (°)	I (°)	k	α_{95} (°)	N	D (°)	I (°)	k	α_{95} (°)	lat. S (°)	long. E (°)	$d\psi$ (°)	$d\chi$ (°)	symbol	
1 Agua Burros, andesite	6313/22	10	354	−59	6	21	9	31	−59	311	3	65	222	3	5	△	–
2 Agua Burros, basalt	6303/12	10	41	−67	2	57	10	42	−51	91	5	55	209	5	7	□	232 ± 1(
3 Agua Burros, ignimbrite	6293/6302	10	23	−50	61	6	10	24	−48	116	4	69	192	3	5	▽	236 ± 1(
4 Agua Burros, basalt	6283/92	10	241	−22	2	45	5	355	−51	10	25	85	59	23	34	⌓	232 ± 1(
5 Agua Burros, ignimbrite	6273/82	10	182	78	3	35	8	152	71	8	18	62	327	27	31	◆	230 ± 1(
6 Rio Atuel, ignimbrite	6323/32	10	136	61	9	17	8	194	68	75	6	71	264	8	10	●	228 ± 1(

samples from each unit are shown in Fig. 5. Note that two representative a.c. demagnetization curves are shown for unit 2. It can be seen by the shapes of these curves that a wide spectrum of remanent coercivities is represented. The best demagnetizing field strength for the samples from the units 2, 3, 4 and 6 is 150 Oe and for the units 1 and 5 oscillates between 150 and 200 Oe.

We isolate stable remanence in only 5 samples from unit 4 (Table 3). The two oldest units (5 and 6) yield directions toward the south with positive inclinations; these units are reversed with respect to the present field direction. The youngest units (1, 2, 3 and 4) yield directions toward the north with negative inclinations. Directions of the stable remanence of samples from each unit are well grouped. Particularly, directions of the stable remanence of samples from units 1, 2, 4 and 6 are substantially better grouped after a.c. demagnetization than before (Table 3). The mean directions of magnetization for each unit, referred to the palaeohorizontal, after a.c. demagnetization, are given in Table 3 and plotted in Fig. 6. The occurrence of well-grouped directions, divergent from the present geomagnetic field, and the presence of a reversal means that substantial secondary remanences are absent.

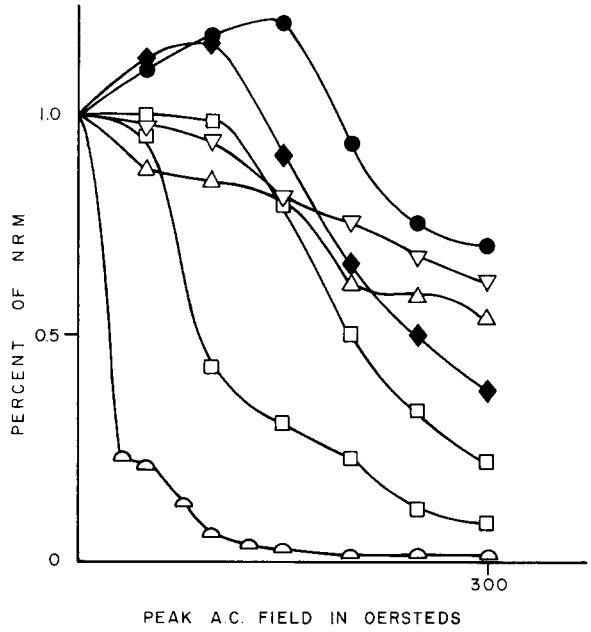


Fig. 5. Alternating-field partial demagnetization curves of representative samples from each palaeomagnetic unit in the Puesto Viejo Formation. Symbols as in Table 3.

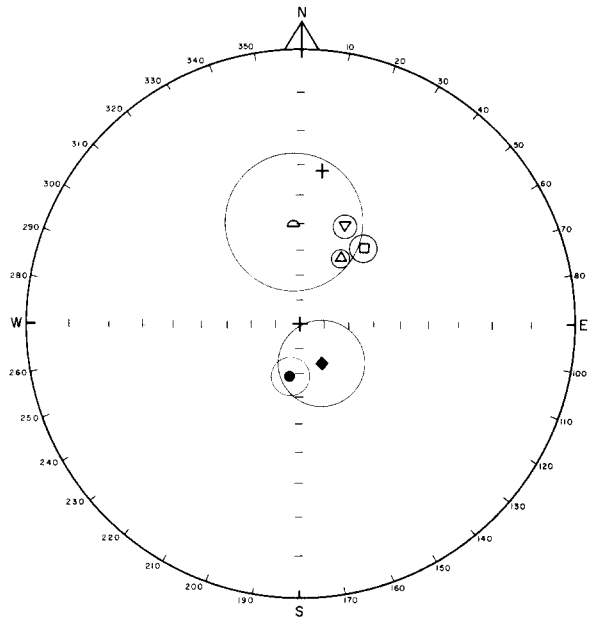


Fig. 6. Site-mean directions of cleaned remanence for the lava flows from the Puesto Viejo Formation to the palaeohorizontal (Table 3). Circles of 95% confidence are shown around each site-mean direction. + = geomagnetic field.

5. Interpretation of the results

Normal and reversed stable remanences have been found in samples from the igneous units of the lower section of the Puesto Viejo Formation. The oldest lava flow at Agua de los Burros shows reversed cleaned remanence; the four igneous flows which rest on it show remanence of normal polarity. Therefore, a reversal of the geomagnetic field (reversed to normal) has been recorded in the igneous units from Agua de los Burros. The ignimbrite from the Rio Atuel Canyon shows reversed stable remanence; this unit could be correlated with the oldest lava flow (ignimbrite) of the Agua de los Burros but the correlation is not conclusive. Therefore, at least one reversal of the geomagnetic field has been recorded in the igneous units of the lower section of the Puesto Viejo Formation. Fossil assemblages occurring at the Puesto Viejo Formation suggest a Lower Triassic age for the upper section of this formation. The mean K–Ar age for five lava flows of the lower section of the Puesto Viejo Formation (232 ± 4 m.y.) indicates a Lower Triassic age. Therefore, a reversal of the geo-

magnetic field has been found in South American rocks of well determined Lower Triassic age. This reversal is within the Illawarra Zone which limits the Late Palaeozoic Interval of reversed geomagnetic polarity (Kiaman Magnetic Interval) at the top [19,1]. Therefore the end of the Kiaman Magnetic Interval has been recognized in South American rocks on the basis of palaeomagnetic and radiometric data. These data indicate, for the first time, that the Illawarra Zone extends at least down to 232 ± 4 m.y. within the Lower Triassic.

At least one reversal of the geomagnetic field has also been found in the Middle Triassic (224 ± 5 m.y.) igneous rocks from Loma de la Estancia in the Ischigualasto-Ischichuca Basin. Particularly, the normal magnetic remanence found in some of these igneous rocks could be compared with the normal remanence found in the Manicovagan Dacite (225 ± 3 m.y.) in Canada [3]. However, it is not possible to suggest a strict correlation of these normal events given that the shortest event of polarity of the geomagnetic field is much smaller than the uncertainty of the radiometric age involved with Triassic rocks.

Virtual geomagnetic poles were computed for each geological unit of the Ischigualasto-Ischichuca Basin and Puesto Viejo Formation (Tables 2 and 3, respectively). Palaeomagnetic poles for these formations may be computed as the mean of these virtual geomagnetic poles (entries 1, $SATr_2$, and 3, $SATr_3$, Table 4) or from the mean directions of their stable magnetic remanences (entries 2 and 4, Table 4). In this table is also given the position of the pole computed as the mean of the virtual geomagnetic poles for the Ischigualasto-Ischichuca Basin and Puesto Viejo Formation (SATr, entry 5). Fig. 7 shows the positions of $SATr_2$ and $SATr_3$; they are reasonably close to the positions of the South

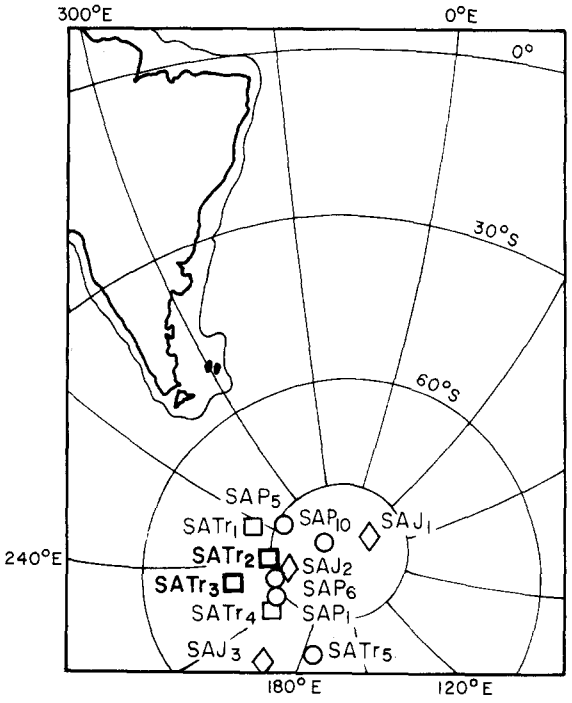


Fig. 7. Middle and Upper Permian (○), Triassic (□) and Jurassic (◇), palaeomagnetic poles for South America. $SATr_2$ and $SATr_3$ are the poles for the igneous rocks from the Ischigualasto-Ischichuca Basin and Puesto Viejo Formation, respectively. Other references in item 5.

TABLE 4

South palaeomagnetic poles for the Triassic lava flows from the Ischigualasto-Ischichuca Basin and Puesto Viejo Formation, Argentina

	Number of time units or V.G.P.	Mean direction of cleaned remanent magnetization				South palaeomagnetic pole						Name
		D (°)	I (°)	k	α_{95} (°)	lat. S (°)	long. E (°)	k	α_{95} (°)	$d\psi$ (°)	dx (°)	
1 Ischigualasto-Ischichuca Basin	9	–	–	–	–	79.2	239.2	13.0	14.8	–	–	$SATr_2$
2 Ischigualasto-Ischichuca Basin	9	189.2	52.1	14.3	14.0	81.7	222.2	–	–	16	20	–
3 Puesto Viejo Formation	6	–	–	–	–	75.9	236.2	14.3	18.2	–	–	$SATr_3$
4 Puesto Viejo Formation	6	16.7	−59.8	27.0	13.1	75.3	230.8	–	–	12	19	–
5 Ischigualasto-Puesto Viejo	15	–	–	–	–	77.9	237.8	14.4	10.4	–	–	SATr
6 Ischigualasto-Puesto Viejo (2, 4)	2	–	–	–	–	78.5	227.6	300.7	14.4	–	–	–
7 Ischigualasto-Puesto Viejo (1, 3)	2	–	–	–	–	77.5	237.5	1161.7	7.3	–	–	–

American palaeomagnetic poles of Triassic age ($SATr_1$, 204 ± 3 m.y. [20,1] and $SATr_4$ [5]). Note that the palaeomagnetic pole for the Upper Section of the Paganzo Group ($SATr_5$, [21]) is here assigned tentatively to the Permian. That is supported by (1) the Upper Carboniferous and Permo-Carboniferous ages suggested for rocks from the Middle Section of the Paganzo Group [22], and (2) the Lower Triassic (232 ± 4 m.y.) age assigned to the Puesto Viejo Formation (it is known that this formation is younger than the Upper Section of the Paganzo Group). The Triassic palaeomagnetic poles of South America form an "age group"; the mean pole position of this group is: 77°S, 239°E ($N = 4$, $\alpha_{95} = 6.6°$, $k = 190$). Fig. 7 also shows that the Triassic, Upper Permian (SAP_1 [21], SAP_{10} [23] and $SATr_5$ [21]) and Middle Permian (SAP_5 [13], SAP_6 [14], 263 ± m.y. [1]) palaeomagnetic poles for South America form a "time group" [24]; the mean pole position of this "time group" is: 80°S, 233°E ($N = 9$, $\alpha_{95} = 4{,}5°$, $k = 126$). The palaeomagnetic poles computed for the Middle Jurassic (161 m.y.) Chon Aike Formation are also shown in Fig. 7 (SAJ_1, [25], SAJ_2 [26] and SAJ_3 [27]). They would form a "time group" with the Middle and Upper Permian and Triassic poles of South America; the mean pole position of this Middle Permian/Middle Jurassic "time group" is: 81°S, 227°E ($N = 12$, $\alpha_{95} = 4.7°$ $k = 84$). Vilas [27] has suggested that the most representative Middle Jurassic pole position for South America is that obtained as the mean of all the virtual geomagnetic poles of the Chon-Aike Formation; in this case, the mean pole position of the South American Middle Permian/Middle Jurassic "time group" is: 81°S, 232°E ($N = 10$, $\alpha_{95} = 4°$, $k = 131$). Therefore, the positions of the Permian, Triassic and Jurassic paleomagnetic poles for South America (including those ones for the Ischigualasto-Ischichuca Basin and Puesto Viejo igneous rocks) support the interpretation that South America had a quasi-

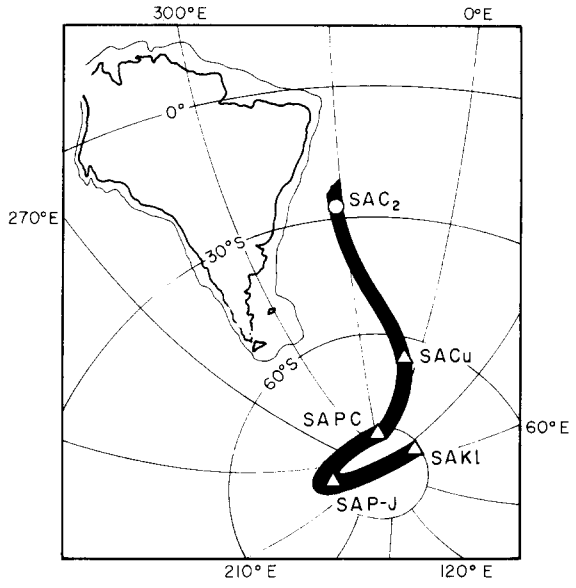


Fig. 8. Late Palaeozoic/Lower Cretaceous section of the polar wandering curve for South America; SACu, SAPC and SAK1 are the mean polar positions for the "age groups" of Upper Carboniferous, Permo-Carboniferous and Lower Cretaceous age, respectively. SAP-J is the mean of the Permo-Jurassic "time group" (Table 5).

TABLE 5

Mean pole positions of the "time" and "age groups" formed by the Late Palaeozoic and Mesozoic paleomagnetic poles of South America

Group	Mean pole position					Name	Palaeomagnetic poles included and references
	lat. S (°)	long. E (°)	N	α_{95} (°)	k		
Lower Cretaceous	82	36	4	7	178	SAK1	Serra Geral [3], Cerro Colorado [31], Rumipalla [32], Los Molinos [33]
Permo-Jurassic	81	232	10	4	131	SAP-J	item 5
Permo-Carboniferous	82	335	4	9	112	SAPC	La Colina (Los Colorados [2], Paganzo and Chaschuil [34,22])
Upper Carboniferous	62	353	5	4	365	SACu	Piaui [30], Paganzo Group (La Rioja [28], Huaco [35,22], and Los Colorados [35,22]), Tubarão [23]

static interval [24] in Middle Permian/Middle Jurassic times.

Fig. 8 shows the Upper Carboniferous/Lower Cretaceous section of the polar wandering curve for South America. SAC_2 is the palaeomagnetic pole for the Carboniferous Taiguati Formation from Bolivia [28]; recently, a Moscovian age has been suggested for this formation [29]. SACu, SAPC and SAK1 are the mean pole positions of the "age groups" formed by the Upper Carboniferous, Permo-Carboniferous and Lower Cretaceous paleomagnetic poles, respectively, and SAP-J is the mean pole position of the Middle Permian/Middle Jurassic "time group" (Table 5). Note the rapid polar shift relative to South America which spanned the time from the Upper Carboniferous (Moscovian) to the Middle Permian. In this episode that continent moved away from the South pole reaching latitudes close to the present ones in the Middle Permian [5]. In this time such movement would have stopped and a quasi-static interval would have started for South America which lasted up to the Middle Jurassic. A new polar shift relative to South America started in Jurassic time; Fig. 8 shows the Jurassic/Early Cretaceous section of this shift which is associated with the formation of the South Atlantic Ocean.

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